GM UNIVERSITY

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Faculty of Engineering and Technology
School of Computer Science and Technology

Regression Analysis(5.1,5.2,5.3)

What is Regression Analysis?

- Regression analysis is a set of statistical methods used for the estimation of relationships between a dependent variable and one or more independent variables. It can be utilized to assess the strength of the relationship between variables and for modeling the future relationship between them.

- For example: Y=F(X)
- HERE, Y IS CALLED DEPENDENT VARIABLE AND X IS CALLED INDEPENDENT VARIABLE

Focus of Regression Analysis

- What is the relationship between the variables?
- What is the strength of the relationships?
- What is the nature of the relationship such as linear or non-linear?
- What is the relevance of the attributes?
- What is the contribution of each attribute?

Applications of Regression Analysis

There are many applications of regression analysis. Some of the applications of regressions include predicting:

- Sales of a goods or services
- Value of bonds in portfolio management
- Premium on insurance companies
- Yield of crops in agriculture
- Prices of real estate

Regression and Correlation

- Correlation quantifies how strong and in which direction the two variables are related. It is being measured using the correlation coefficient, which goes to the extent of negative one and positive one.

- Positive Correlation: In this case, if the value of one of them goes up, then the value of the other one also goes up.
- Negative Correlation: That is, when one variable is high, the other is low, and vice versa.
- No Correlation: Not statistically significantly correlated: partners, financial wealth, and life satisfaction.

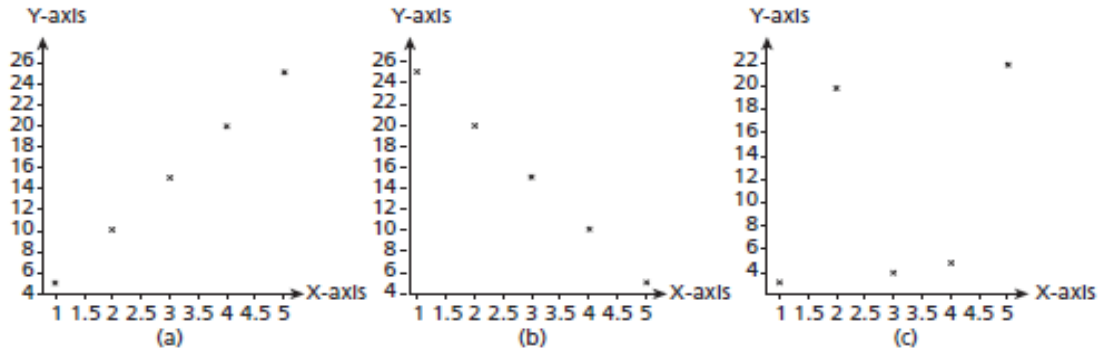


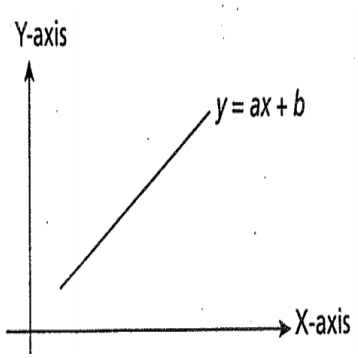
Figure 5.1: Examples of (a) Positive Correlation (b) Negative Correlation (c) Random Points with No Correlation

Regression and Causation

- Causation means one variable directly affects another (cause → effect).
- Correlation does NOT imply causation.
- Just because two variables are related does not mean one causes the other.
- Examples:
- If x causes y, then a change in x leads to a change in y.
- But: Higher temperature and cold drink sales may be related,
- However, other factors (marketing, season, location) may also influence sales.

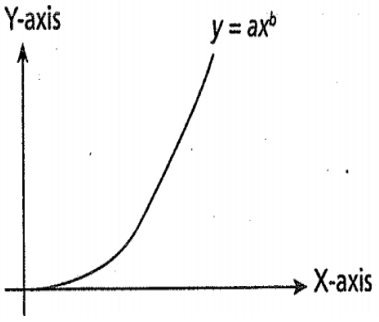
Linear and Non-Linear Relationships

- A. Linear Relationship
- A relationship is linear if it forms a straight line.
- General form:
- y=ax+b
- When x increases, y increases (or decreases) at a constant rate.
- Example:
- Salary increases steadily with years of experience.



Non-Linear Relationship

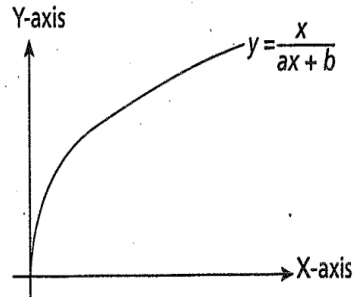
A relationship is non-linear if it is not a straight line.
The rate of change is not constant.
Common type:
1. Exponential Function: y=a^b
Growth becomes faster over time.
Example:
Population growth, compound interest



2. Power Function

$$y = \frac{x}{ax+b}$$

Curve increases but slows down.
Example:
Efficiency or saturation effects.



Types of Regression

- **Linear Regression**

It is a type of regression where a line is fitted upon given data for finding the linear relationship between one independent variable and one dependent variable to describe relationships.

- **Multiple Regression**

It is a type of regression where a line is fitted for finding the linear relationship between two or more independent variables and one dependent variable to describe relationships among variables.

- **Polynomial Regression**

It is a type of non-linear regression method of describing relationships among variables where Nth degree polynomial is used to model the relationship between one independent variable and one dependent variable. Polynomial multiple regression is used to model two or more independent variables and one dependent variable.

- **Logistic Regression**

It is used for predicting categorical variables that involve one or more independent variables and one dependent variable. This is also known as a binary classifier.

- **Lasso and Ridge Regression Methods**

These are special variants of regression method where regularization methods are used to limit the number and size of coefficients of the independent variables.

Types of Regression Methods

The classification of regression methods is shown in Figure 5.3.

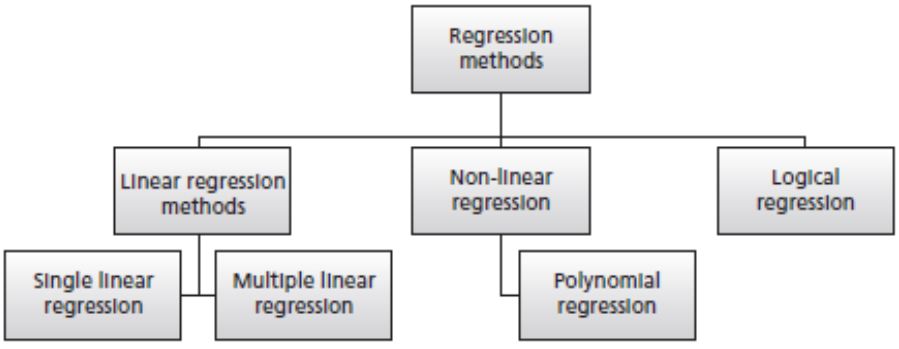


Figure 5.3: Types of Regression Methods

Limitations of Regression Method

- **Outliers –** Outliers are abnormal data. It can bias the outcome of the regression model, as outliers push the regression line towards it.
- **Number of cases –** The ratio of independent and dependent variables should be at least 20 : 1. For every explanatory variable, there should be at least 20 samples. Atleast five samples are required in extreme cases.
- **Missing data –** Missing training data can make the model unfit for the sampled data.
- **Multicollinearity –** If exploratory variables are highly correlated (0.9 and above), the regression is vulnerable to bias. Singularity leads to perfect correlation of 1. The remedy is to remove exploratory variables that exhibit correlation more than 1. If there is a tie, then the tolerance (1 − R squared) is used to eliminate variables that have the greatest value.

Linear Regression

- Linear Regression is a fundamental supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables. It predicts continuous values by fitting a straight line that best represents the data.
- It assumes that there is a linear relationship between the input and output
- Uses a best-fit line to make predictions
- Commonly used in forecasting, trend analysis, and predictive modelling

Linear Regression (Continued)

- The assumptions of linear regression are listed as follows:
- The observations (y) are random and are mutually independent.
- The difference between the predicted and true values is called an error. The error is also mutually independent with the same distributions such as normal distribution with zero mean and constant variables.
- The distribution of the error term is independent of the joint distribution of explanatory variables.
- The unknown parameters of the regression models are constants.

$$y = a_0 + a_1 \times x + e$$

Linear Regression Model

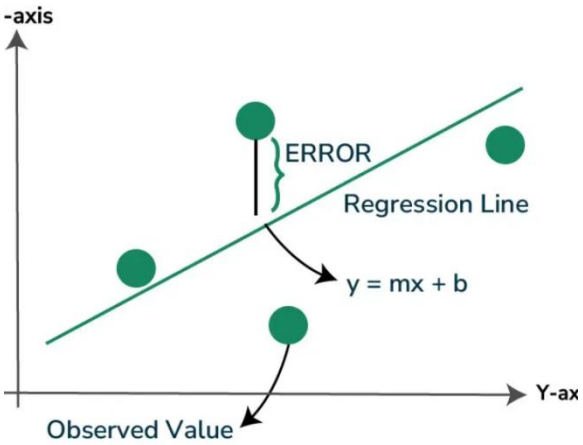
$$J(a_1, a_0) = \sum_{i=1}^{n}[y_i - f(x_i)]^2$$

$$= \sum_{i=1}^{n}[y_i - (a_0 + a_1 x_i)]^2$$

$$a_1 = \frac{(\overline{xy}) - (\overline{x})(\overline{y})}{(\overline{x^2}) - (\overline{x})^2} \quad (5.9)$$

And the value of a_0 is given as follows:

$$a_0 = (\overline{y}) - a_1 \times \overline{x} \quad (5.10)$$



Linear Regression Example

Example 5.1: Let us consider an example where the five weeks' sales data (in Thousands) is given as shown below in Table 5.1. Apply linear regression technique to predict the 7th and 9th month sales.

Table 5.1: Sample Data

x_i (Week)	y_i (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

Table 5.2: Computation Table

x_i	y_i	$(x_i)^2$	$x_i \times y_i$
1	1.2	1	1.2
2	1.8	4	3.6
3	2.6	9	7.8
4	3.2	16	12.8
5	3.8	25	19
Sum = 15 Average of (x_i) $= \overline{x} = \frac{15}{5}$ $= 3$	Sum = 12.6 Average of (y_i) $= \overline{y} = \frac{12.6}{5}$ $= 2.52$	Sum = 55 Average of (x_i^2) $= \overline{x_i^2} = \frac{55}{5}$ $= 11$	Sum = 44.4 Average of $(x_i \times y_i)$ $= \overline{xy} = \frac{44.4}{5}$ $= 8.88$

Let us compute the slope and intercept now using Eq. (5.9) as:

$$a_1 = \frac{8.88 - 3(2.52)}{11 - 3^2} = 0.66$$

$$a_0 = 2.52 - 0.66 \times 3 = 0.54$$

The fitted line is shown in Figure 5.5.

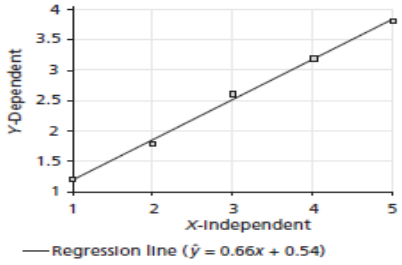


Figure 5.5: Linear Regression Model Constructed

Let us model the relationship as $y = a_0 + a_1 \times x$. Therefore, the fitted line for the above data is: $y = 0.54 + 0.66 \times x$.

The predicted 7th week sale would be (when $x = 7$), $y = 0.54 + 0.66 \times 7 = 5.16$ and the 12th month, $y = 0.54 + 0.66 \times 12 = 8.46$. All sales are in thousands.